

RETAIL ANALYTICS

Power BI Project

Project Documentation

Tool	Microsoft Power BI Desktop
Report File	Final_Project.pbix
Data Model Type	Star Schema (3 Dimension + 1 Fact + 1 Returns + Calculation Group)
Date	April 2026

1. Project Planning & Management

1.1 Project Proposal

Overview:

This project is a Retail Analytics Dashboard built in Microsoft Power BI. It ingests transactional retail data and transforms it into actionable intelligence for sales, profitability, returns analysis, and retailer behaviour.

Objectives:

- Provide a unified view of sales performance, profitability, and order return trends.
- Enable time-series comparison (MTD / QTD / YTD / PY) through reusable Calculation Groups.
- Support strategic decision-making via RFM segmentation, discount sensitivity, and geospatial efficiency analysis.
- Deliver interactive What-If simulations so management can test pricing and discount scenarios before committing.

1.2 Project Plan – Timeline & Milestones

Phase	Deliverable	Student Deadline	Graduate Deadline	Status
1 – Planning	Proposal, Gantt chart, KPIs	20 Feb 2026	27 Jan 2026	Completed
2 – Literature Review	Benchmark report, Power BI best practices	20 Apr 2026	1 Mar 2026	Completed
3 – Requirements	Stakeholder needs, User stories, Functional spec	20 Apr 2026	1 Mar 2026	Completed
4 – Design	Data model (Star Schema), DAX architecture, UI mockups	1 May 2026	3 Apr 2026	In Progress
5 – Implementation	PBIX report, DAX measures, Calculation Groups	10 Jul 2026	17 May 2026	In Progress
6 – Testing & Reports	Test cases, final PPT, video demo	17 Jul 2026	24 May 2026	Pending

1.3 Risk Assessment & Mitigation

Risk	Likelihood	Impact	Mitigation
Data quality issues (nulls, wrong types)	Medium	High	Add data-cleaning steps in Power Query (remove nulls, validate data types)
Incorrect DAX DIVIDE causing silent errors	Low	High	Always use DIVIDE() with 0 as alternate result; add data tests
Slow report performance with large dataset	Medium	Medium	Use aggregations, limit row-level visuals, enable query folding
Misinterpretation of Return Rate %	Low	Medium	Add tooltips and info buttons explaining each KPI formula
Scope creep (adding too many pages)	High	Medium	Lock page scope in planning; create backlog for v2 features

1.4 KPIs (Key Performance Indicators)

KPI Name	DAX Formula	Target / Threshold
Total Sales	SUM(F_Orders[Sales_Amount])	Track MoM growth
Total Profit	SUM(F_Orders[Profit])	> 0 always
Profit Margin %	DIVIDE([Total Profit],[Total Sales],0)	> 15%
Return Rate %	DIVIDE([Returned Orders],[Total Orders],0)	< 10%
Net Sales	[Total Sales] * (1 - [Return Rate %])	Primary revenue KPI
Average Order Value (AOV)	DIVIDE([Total Sales],[Total Orders],0)	Trend upward QoQ
Total Quantity	SUM(F_Orders[Order_Quantity])	Volume benchmark

2. Literature Review

The following concepts and best practices informed the design of this Power BI project:

2.1 Star Schema in Business Intelligence

A Star Schema separates transactional facts from descriptive dimensions, enabling fast aggregation and simple DAX. This project adopts a star schema with F_Orders as the central fact table surrounded by D_Products, D_Retailers, and D_Date dimensions, plus a separate D>Returns table for cleaner return-rate logic.

2.2 DAX Calculation Groups for Time Intelligence

Microsoft introduced Calculation Groups to eliminate DAX measure proliferation. A single Calculation Group can apply MTD, QTD, YTD, and Previous Year transformations across all base measures dynamically. This project uses a 'Calculation group' table for exactly this purpose, enabling users to switch time context without creating separate YTD_Sales, YTD_Profit, etc. measures.

2.3 RFM Analysis in Retail

Recency-Frequency-Monetary (RFM) segmentation classifies customers (or retailers) into behavioural cohorts. Champions show high recency, frequency, and monetary value, while Hibernating retailers have not ordered recently. Applying RFM to D_Retailers allows the business to prioritise outreach and tailor promotional strategies by segment.

2.4 What-If Analysis in Power BI

Power BI's What-If Parameters create a numeric slicer backed by a disconnected table. Combined with DAX measures that reference the slicer value, these enable real-time scenario modelling, such as simulating the profit impact of increasing or decreasing product discounts.

2.5 Discount Sensitivity & Price Elasticity

Marketing research shows discount effectiveness follows an S-curve: small discounts have negligible volume impact, moderate discounts hit the 'sweet spot' driving volume without eroding margin, and large discounts become margin-negative. Analysing F_Orders[Product_Discount] against Net Sales by Product_Category surfaces this curve for each category.

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Role	Key Need
Sales Manager	Primary user	Real-time sales & profit KPIs by region, product, and period
Operations Manager	Secondary user	Return rate trends, distance-vs-return analysis, logistics cost insights
Finance Team	Reviewer	Accurate Profit Margin % and Net Sales figures with time comparisons
Marketing Team	Analyst	Discount sensitivity curves, RFM segments for targeted campaigns
Executive Leadership	Dashboard consumer	High-level KPI summary with YoY growth and What-If scenarios

3.2 Functional Requirements

- **FR-01: Display Total Sales, Total Profit, Net Sales, Return Rate %, AOV, and Profit Margin % as card KPIs.**
- **FR-02: Allow time filtering via Calculation Group (MTD, QTD, YTD, Previous Year, Growth %).**
- **FR-03: Show Scatter Plot of Distance_from_Warehouse vs Return Rate % per retailer.**
- **FR-04: Render a Product Quadrant matrix classifying products by Profit Margin % and Return Rate %.**
- **FR-05: Display Discount Sensitivity chart (Product_Discount vs Net Sales by Product_Category).**
- **FR-06: Provide an RFM segmentation table for D_Retailers (Champion / Loyal / At-Risk / Hibernating).**
- **FR-07: Include a What-If slider for Discount % showing simulated profit impact in real time.**
- **FR-08: Enable cross-filtering between all visuals on the same page.**

3.3 Non-Functional Requirements

- **NFR-01 Performance: Dashboard loads in under 5 seconds on standard hardware.**
- **NFR-02 Usability: All KPI cards include tooltips explaining the DAX formula.**
- **NFR-03 Reliability: All measures use DIVIDE() to avoid divide-by-zero errors.**
- **NFR-04 Maintainability: Measures centralised in _Measures table; no inline calculations in visuals.**
- **NFR-05 Scalability: Data model supports incremental refresh for growing datasets.**

4. System Analysis & Design

4.1 Data Model – Star Schema

The data model follows a Star Schema pattern. F_Orders is the central fact table. Three dimension tables (D_Date, D_Products, D_Retailers) and one return-tracking table (D>Returns) surround it. A dedicated _Measures table holds all DAX measures, and a Calculation Group table powers dynamic time intelligence.

4.2 Tables & Columns

F_Orders (Fact Table)

Column	Type	Description
Order_ID	Text / Key	Unique order identifier – used in DISTINCTCOUNT for Total Orders
Sales_Amount	Decimal	Revenue per order line – basis for Total Sales and Net Sales
Profit	Decimal	Profit per order line – basis for Total Profit and Profit Margin %
Order_Quantity	Integer	Units ordered – basis for Total Quantity
Product_Price	Decimal	Unit price before discount
Product_Discount	Decimal (0-1)	Discount fraction applied to the order
Product_Category	Text	Product category – used for discount sensitivity segmentation

D_Retailers (Dimension Table)

Column	Type	Description
Retailer_ID	Text / Key	Primary key – joins to F_Orders
Retailer_Name	Text	Retailer display name
Region / City	Text	Geographic attributes for spatial filtering
Distance_from_Warehouse	Decimal (km)	Key for Geospatial Efficiency Analysis – correlates with Return Rate %

D_Products (Dimension Table)

Column	Type	Description
Product_ID	Text / Key	Primary key – joins to F_Orders
Product_Name	Text	Product display name
Product_Category	Text	Category for grouping (Discount Sensitivity Analysis)
Product_SubCategory	Text	Sub-category for drill-down analysis

D_Date (Date Dimension)

Column	Type	Description
Date	Date / Key	Daily grain date key – joins to F_Orders order date
Year / Quarter / Month	Integer	Hierarchy fields for time-series visuals
Week_Number	Integer	Weekly aggregation
Is_Weekend	Boolean	Flag for weekday vs weekend segmentation

D_Returns (Return Tracking Table)

Column	Type	Description
Order_ID	Text / Key	Foreign key – joins to F_Orders; drives COUNTROWS and DISTINCTCOUNT for return measures
Return_Date	Date	Date the return was registered
Return_Reason	Text	Optional reason code for future root-cause analysis

4.3 DAX Measures (_Measures Table)

Measure	DAX Formula	Business Meaning
Total Sales	SUM(F_Orders[Sales_Amount])	Gross revenue before returns
Total Profit	SUM(F_Orders[Profit])	Gross profit before returns
Total Quantity	SUM(F_Orders[Order_Quantity])	Units sold
Total Orders	DISTINCTCOUNT(F_Orders[Order_ID])	Distinct order count
Total Returns	COUNTROWS(D_Returns)	All return records
Returned Orders	DISTINCTCOUNT(D_Returns[Order_ID])	Orders with at least one return
Return Rate %	DIVIDE([Returned Orders],[Total Orders],0)	% of orders returned
Net Sales	[Total Sales] * (1 - [Return Rate %])	Revenue adjusted for returns
AOV	DIVIDE([Total Sales],[Total Orders],0)	Average revenue per order
Profit Margin %	DIVIDE([Total Profit],[Total Sales],0)	Profit as % of revenue

4.4 Calculation Group – Time Intelligence

The 'Calculation group' table provides the following dynamic Calculation Items, applied to any selected base measure:

Calculation Item	Purpose	DAX Pattern
Current Period	No transformation – returns base measure value	SELECTEDMEASURE()
MTD	Month-to-date accumulation	CALCULATE(SELECTEDMEASURE(), DATESMTD(D_Date[Date]))
QTD	Quarter-to-date accumulation	CALCULATE(SELECTEDMEASURE(), DATESQTD(D_Date[Date]))
YTD	Year-to-date accumulation	CALCULATE(SELECTEDMEASURE(), DATESYTD(D_Date[Date]))
Previous Year	Same period last year	CALCULATE(SELECTEDMEASURE(), SAMEPERIODLASTYEAR(D_Date[Date]))
Growth % (Planned)	YoY absolute or % change via new Calculation Item	SELECTEDMEASURE() - CALCULATE(SELECTEDMEASURE(), SAMEPERIODLASTYEAR(...))

4.5 Planned Analyses (Advanced Features)

Analysis 1 – Geospatial Efficiency (Distance vs Return Rate)

Visual: Scatter Plot

- X-axis: D_Retailers[Distance_from_Warehouse]

- Y-axis: [Return Rate %]
- Size/Colour: [Total Sales] per retailer

Hypothesis: Retailers far from the warehouse experience higher return rates due to extended transit time causing product damage. The scatter will reveal if a distance threshold exists beyond which return rates spike, guiding decisions on regional warehouse expansion or specialised packaging.

Analysis 2 – Product Quadrant (Toxic Product Matrix)

Visual: Scatter Plot with four quadrant reference lines

- X-axis: [Profit Margin %] per product
- Y-axis: [Return Rate %] per product

Quadrant Classification:

- Golden (high margin, low returns): Priority products to promote
- Toxic (high margin, high returns): Hidden quality issue – investigate root cause
- Burden (low margin, high returns): Candidates for discontinuation
- Acceptable (low margin, low returns): Steady contributors, improve margin

Analysis 3 – Discount Sensitivity

Visual: Line or Area chart

- X-axis: F_Orders[Product_Discount] bucketed (0-5%, 5-10%, 10-15%, >15%)
- Y-axis: [Net Sales] or [Total Quantity]
- Legend: Product_Category

Goal: Identify the sweet spot where discounting drives measurable volume uplift before margin erosion exceeds the benefit. Different categories are expected to have different elasticity curves.

Analysis 4 – RFM Retailer Segmentation

Implemented via a DAX Calculated Table on D_Retailers:

- Recency (R): Days since last order date in F_Orders
- Frequency (F): Count of distinct orders per retailer
- Monetary (M): Total Sales_Amount per retailer

Segments: Champions (R=5,F=5,M=5), Loyal Customers, Potential Loyalists, At-Risk, Hibernating. Allows targeted marketing campaigns and priority support for high-value retailers.

Analysis 5 – What-If Discount Simulator

Implementation:

- Create a What-If Parameter 'Discount Adjustment %' (range -20% to +20%)
- DAX: Simulated Profit = [Total Profit] + ([Total Sales] * 'Discount Adj'[Discount Adj Value])
- Visuals: KPI card showing Simulated Profit vs Actual Profit; gauge showing margin impact

Business value: Executives can drag the slider to instantly see how a global discount change affects the bottom line before a pricing decision is made.

5. Implementation (Power BI Report)

5.1 Technology Stack

Layer	Technology	Details
Reporting	Microsoft Power BI Desktop	Version: CY26SU02 File: Final_Project.pbix
Data Modelling	Tabular Model (Vertipaq)	In-memory columnar engine; Star Schema
DAX Engine	DAX (Data Analysis Expressions)	Measures in _Measures table; Calculation Groups for time intel
ETL / Data Prep	Power Query (M Language)	Data cleaning, type enforcement, relationship keys
Version Control	GitHub	All .pbix and documentation files committed to repository
Documentation	Microsoft Word (.docx)	This document generated from Project Documentation Guidelines

5.2 Data Model Architecture

The report uses a classic Star Schema:

- Centre: F_Orders (fact table with Sales_Amount, Profit, Order_Quantity, Product_Discount, Product_Price)
- Dimensions: D_Date (date grain), D_Products (product attributes), D_Retailers (retailer + distance data)
- Return Table: D>Returns (linked via Order_ID for return rate calculations)
- Measures Table: _Measures (all KPIs centralised – no implicit measures)
- Calculation Group: Provides dynamic MTD/QTD/YTD/PY time context for any selected measure

5.3 Report Pages (Planned Structure)

Page	Purpose	Key Visuals
1 – Executive Summary	Top-level KPI overview	KPI cards: Total Sales, Net Sales, Profit Margin %, Return Rate %; trend line; Calculation Group slicer
2 – Sales Performance	Sales by category, product, time	Bar chart by Product_Category; line chart over time; matrix by retailer
3 – Returns Analysis	Deep-dive on return patterns	Scatter: Distance vs Return Rate; table of top-returned products
4 – Product Quadrant	Toxic product identification	Scatter: Profit Margin % vs Return Rate % with quadrant lines
5 – Discount Sensitivity	Pricing optimisation	Line chart: Discount bucket vs Net Sales by Category
6 – Retailer RFM	Retailer segmentation	Matrix of RFM scores; bar chart of segment counts; map by region
7 – What-If Simulator	Scenario modelling	What-If slider; Simulated vs Actual Profit KPI cards; gauge

5.4 Coding Standards & Best Practices

- All measures stored in _Measures table (no implicit/auto measures).
- DIVIDE() used for every division to prevent blank/error results.
- Calculation Group used for time intelligence to avoid measure proliferation.
- Power Query steps named descriptively (e.g., 'Remove Null Order IDs', 'Enforce Decimal for Sales_Amount').
- Visual-level filters avoided; use report or page-level filters for consistency.
- Relationship direction: Single (one-to-many from dimension to fact).

6. Testing & Quality Assurance

6.1 Test Cases

Test ID	Test Description	Expected Result	Status
TC-01	Total Sales equals SUM of all Sales_Amount rows	Matches manual Excel SUM	Pass
TC-02	Return Rate % = 0 when D_Returns is empty	Measure returns 0 (not blank)	Pass
TC-03	Net Sales < Total Sales when Return Rate % > 0	Arithmetic verified	Pass
TC-04	Calculation Group YTD accumulates correctly across months	Jan+Feb+Mar sum = Mar YTD	Pass
TC-05	Distance scatter plot shows correct retailer count	Count matches D_Retailers row count	Pending
TC-06	What-If slider at 0% change returns Simulated = Actual Profit	Delta = 0	Pending
TC-07	RFM segmentation covers 100% of retailers with no nulls	No unclassified retailers	Pending
TC-08	Cross-filter: selecting one product filters all page visuals	All visuals reflect selection	Pending

6.2 Known Issues / Bug Log

Issue ID	Description	Resolution
BUG-01	Order_Quantity column has a leading space: ' Order_Quantity ' causing merge errors	Rename column in Power Query: Table.RenameColumns
BUG-02	Return Rate % may exceed 100% if Order_ID duplicates exist in F_Orders	Add deduplication step in Power Query for F_Orders[Order_ID]

7. Final Presentation & Reports

7.1 Deliverables Checklist

Deliverable	Format	Status
Project Proposal & Plan	Word (.docx)	Completed
Data Model Documentation (this document)	Word (.docx)	Completed
Final Power BI Report	PBIX (Final_Project.pbix)	In Progress
DAX Measures Reference Sheet	Included in Section 5 above	Completed
Test Cases & Bug Log	Included in Section 6 above	Completed
Project Presentation Slides	PPT / PDF	Pending
User Manual	Word (.docx) or PDF	Pending
Video Demonstration (Optional)	MP4 / YouTube link	Pending

7.2 User Manual Summary

How to use the Final_Project.pbix report:

- Open the file in Power BI Desktop (version CY26SU02 or later recommended).
- Use the Calculation Group slicer on any page to switch between Current, MTD, QTD, YTD, and Previous Year views.
- Use the What-If Discount Adjustment slider (Page 7) to simulate pricing changes.
- Click any bar, data point, or product on one visual to cross-filter all other visuals on the page.
- Hover over any KPI card to see the DAX formula tooltip.

7.3 Future Enhancements (v2 Backlog)

- Add 'Previous Year Comparison' Calculation Item showing both absolute variance and Growth % dynamically.
- Integrate a live data source (e.g., SQL Server or SharePoint) using DirectQuery or scheduled refresh.
- Add AI-powered insights via Power BI's Smart Narrative visual for automatic commentary.
- Implement Row-Level Security (RLS) so each regional manager sees only their territory's data.
- Build a mobile-optimised layout for the Executive Summary page.

All documents and the PBIX file are to be uploaded to the project GitHub repository as per the Project Documentation Guidelines.